

# 周晟炜

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## 研究兴趣

公平分配、算法博弈论、机制设计、在线算法、近似算法。

## 教育背景

- 2021 – 2025 博士, 计算机科学, 澳门大学, 澳门特别行政区  
博士论文: *Fairly Allocating Indivisible Chores to Symmetric and Asymmetric Agents*  
导师: 吴晓伟教授
- 2019 – 2020 硕士, 城市信息学, 伦敦国王学院, 英国伦敦  
论文导师: Angus Roberts 博士
- 2014 – 2019 学士, 城乡规划, 武汉大学, 中国武汉  
辅修: 地理信息科学

## 研究经历

- 2025 – 至今 博士后研究员, 南洋理工大学, 新加坡  
导师: 贝小辉教授
- 2021 研究助理, 澳门大学, 澳门特别行政区  
导师: 吴晓伟教授

## 荣誉与奖学金

- 2025 最佳学生论文奖, *Conference on Web and Internet Economics (WINE)*  
225 篇投稿中唯一获奖论文
- 2025 太湖奖学金, 澳门大学  
粤港澳大湾区太湖奖学金
- 2021 – 2025 澳门大学博士助学金, 澳门大学
- 2019 优秀学士学位论文奖, 武汉大学  
本科毕业论文获评城市设计学院前 5%

## 工作论文 $(\alpha-\beta)$ : 作者按字母序    ✉: 通讯作者

- [W5] **Subsidised Proportionality in Fair Allocation**  
 $(\alpha-\beta)$  Xiaowei Wu, Cong Zhang, Shengwei Zhou<sup>✉</sup>.  
*Operations Research* 小修 (minor revision)
- [W4] **Degree-bounded Online Bipartite Matching Revisited: OCS vs. Ranking**  
 $(\alpha-\beta)$  Yilong Feng, Haolong Li, Xiaowei Wu, Shengwei Zhou.  
投稿至 *Mathematics of Operations Research*

- [W3]    **Efficient EFX Allocations for Chores**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*Artificial Intelligence* 在审
- [W2]    **An FPTAS for 7/9-Approximation to Maximin Share Allocations**  
 $(\alpha\text{-}\beta)$  Xin Huang, Shengwei Zhou.  
 工作论文
- [W1]    **When Maximum Nash Welfare Becomes Strongly Fair: Bi-valued Goods**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
 工作论文

### 会议论文    $(\alpha\text{-}\beta)$ : 作者按字母序

- [C17]    **Fair Division by Contribution: A Shapley Value Perspective**  
 $(\alpha\text{-}\beta)$  Xiaohui Bei, Pinyan Lu, Xiaowei Wu, Shengwei Zhou.  
*Proceedings of the 27th ACM Conference on Economics and Computation (EC)*, 2026, Forthcoming.
- [C16]    **Allocating Chores with Restricted Additive Costs: Achieving EFX, MMS, and Efficiency Simultaneously**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*Proceedings of the ACM on Web Conference (WWW)*, pages 146-156, Apr 2026.  
 doi:[10.1145/3774904.3792300](https://doi.org/10.1145/3774904.3792300)
- [C15]    **All-but-One MMS Allocation for Chores**  
 $(\alpha\text{-}\beta)$  Jiawei Qiu, Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*Proceedings of the ACM on Web Conference (WWW)*, pages 157-168, Apr 2026.  
 doi:[10.1145/3774904.3792305](https://doi.org/10.1145/3774904.3792305)
- [C14]    **Degree-bounded Online Bipartite Matching Revisited: OCS vs. Ranking**  
 $(\alpha\text{-}\beta)$  Yilong Feng, Haolong Li, Xiaowei Wu, Shengwei Zhou.  
*Proceedings of the 21st Conference on Web and Internet Economics (WINE)*, pages 747–764, Dec 2025. doi:[10.1007/978-3-032-18660-7\\_39](https://doi.org/10.1007/978-3-032-18660-7_39) ★ 最佳学生论文
- [C13]    **Incentive Analysis of Collusion in Fair Division**  
 $(\alpha\text{-}\beta)$  Haoqiang Huang, Biaoshuai Tao, Mingwei Yang, Shengwei Zhou.  
*Proceedings of the 21st Conference on Web and Internet Economics (WINE)*, pages 392–410, Dec 2025. doi:[10.1007/978-3-032-18660-7\\_21](https://doi.org/10.1007/978-3-032-18660-7_21)
- [C12]    **When is Truthfully Allocating Chores no Harder than Goods?**  
 $(\alpha\text{-}\beta)$  Bo Li, Biaoshuai Tao, Fangxiao Wang, Xiaowei Wu, Mingwei Yang, Shengwei Zhou.  
*Proceedings of the 18th International Symposium on Algorithmic Game Theory (SAGT)*, pages 247-264, Sep 2025. doi:[10.1007/978-3-032-03639-1\\_14](https://doi.org/10.1007/978-3-032-03639-1_14).

- [C11] **Approximately EFX and PO Allocations for Bivalued Chores**  
 $(\alpha\text{-}\beta)$  Zehan Lin, Xiaowei Wu, Shengwei Zhou.  
*Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 3952-3960, Aug 2025. doi:[10.24963/ijcai.2025/440](https://doi.org/10.24963/ijcai.2025/440).
- [C10] **A Little Subsidy Ensures MMS Allocation for Three Agents**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Quan Xue, Shengwei Zhou.  
*Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 4073-4081, Aug 2025. doi:[10.24963/ijcai.2025/454](https://doi.org/10.24963/ijcai.2025/454).
- [C9] **Revisiting Proportional Allocation with Subsidy: Simplification and Improvements**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Quan Xue, Shengwei Zhou.  
*Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 4082-4090, Aug 2025. doi:[10.24963/ijcai.2025/455](https://doi.org/10.24963/ijcai.2025/455).
- [C8] **Pure Nash Equilibrium of Weight Picking Sequence Protocol is WEF1 for Two Agents**  
 $(\alpha\text{-}\beta)$  Rufan Bai, Huahua Miao, Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*Proceedings of the 19th International Joint Conference, IJTCS-FAW*, pages 238-251, Jun 2025. doi:[10.1007/978-981-96-8312-3\\_18](https://doi.org/10.1007/978-981-96-8312-3_18).
- [C7] **Tree Splitting Based Rounding Scheme for Weighted Proportional Allocations with Subsidy**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Shengwei Zhou.  
*Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 295-313, Dec 2024. doi:[10.1007/978-3-032-08560-3\\_17](https://doi.org/10.1007/978-3-032-08560-3_17).
- [C6] **On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores**  
 $(\alpha\text{-}\beta)$  Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, Shengwei Zhou.  
*Proceedings of the 18th International Joint Conference, IJTCS-FAW*, pages 33-52, Aug 2024. doi:[10.1007/978-981-97-7752-5\\_3](https://doi.org/10.1007/978-981-97-7752-5_3).
- [C5] **One Quarter Each (on Average) Ensures Proportionality**  
 $(\alpha\text{-}\beta)$  Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 582-599, Dec 2023. doi:[10.1007/978-3-031-48974-7\\_33](https://doi.org/10.1007/978-3-031-48974-7_33).
- [C4] **Improved Competitive Ratio for Edge-Weighted Online Stochastic Matching**  
 Guoliang Qiu, Yilong Feng, Shengwei Zhou, Xiaowei Wu.  
*Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, pages 527-544, Dec 2023. doi:[10.1007/978-3-031-48974-7\\_30](https://doi.org/10.1007/978-3-031-48974-7_30).

- [C3] **Multi-agent Online Scheduling: MMS Allocations for Indivisible Items**  
Shengwei Zhou, Rufan Bai, Xiaowei Wu.  
*Proceedings of the 40th International Conference on Machine Learning (ICML)*,  
pages 42506-42516, Jul 2023. doi:[10.5555/3618408.3620197](https://doi.org/10.5555/3618408.3620197).
- [C2] **Weighted EF1 Allocations for Indivisible Chores**  
( $\alpha$ - $\beta$ ) Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*Proceedings of the 24th ACM Conference on Economics and Computation (EC)*,  
pages 1155, Jul 2023. doi:[10.1145/3580507.3597763](https://doi.org/10.1145/3580507.3597763).
- [C1] **Approximately EFX Allocations for Indivisible Chores**  
Shengwei Zhou, Xiaowei Wu.  
*Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI)*, pages 783-789, Aug 2022. doi:[10.24963/ijcai.2022/110](https://doi.org/10.24963/ijcai.2022/110).

## 期刊论文 ( $\alpha$ - $\beta$ ): 作者按字母序

- [J3] **Weighted EF1 Allocations for Indivisible Chores**  
( $\alpha$ - $\beta$ ) Xiaowei Wu, Cong Zhang, Shengwei Zhou.  
*Artificial Intelligence (AIJ)*, 347:104386, Oct 2025.  
doi:[10.1016/j.artint.2025.104386](https://doi.org/10.1016/j.artint.2025.104386).  
会议版本: EC 23 (C2)。
- [J2] **On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores**  
( $\alpha$ - $\beta$ ) Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, Shengwei Zhou<sup>✉</sup>.  
*Theoretical Computer Science (TCS)*, 1042:115248, Jul 2025.  
doi:[10.1016/j.tcs.2025.115248](https://doi.org/10.1016/j.tcs.2025.115248). 会议版本: IJTCS-FAW 24 (C6)。
- [J1] **Approximately EFX Allocations for Indivisible Chores**  
Shengwei Zhou, Xiaowei Wu.  
*Artificial Intelligence (AIJ)*, 326:104037, Jan 2024.  
doi:[10.1016/j.artint.2023.104037](https://doi.org/10.1016/j.artint.2023.104037).  
会议版本: IJCAI 22 (C1)。

## 教学经历

- 2022 秋  
– 2024 助教, *University of Macau*  
CISC7007 Design and Analysis of Algorithm (研究生课程)
- 2022 春  
– 2025 助教, *University of Macau*  
CISC2006 Algorithm Design and Analysis (本科课程)
- 2021 秋 助教, *University of Macau*  
CISC3027 Special Topics in Computer and Information Science (本科课程)

## 学术报告

|             |                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 年 7 月  | Fair Division by Contribution: A Shapley Value Perspective<br><i>ACM EC, Rome</i>                                                        |
| 2025 年 8 月  | Revisiting Proportional Allocation with Subsidy: Simplification and Improvements<br><i>IJCAI, Guangzhou</i>                              |
| 2025 年 4 月  | Revisiting Proportional Allocation with Subsidy: Simplification and Improvements<br><i>IOTSC Postgraduate Forum, University of Macau</i> |
| 2024 年 10 月 | Fair Allocations of Chores with Subsidy<br><i>Invited Talk, Kyushu University</i>                                                        |
| 2024 年 8 月  | On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores<br><i>IJTCS-FAW, Hong Kong</i>                                |
| 2024 年 3 月  | PROP Allocation for Indivisible Chores with Subsidy<br><i>Invited Talk, Shanghai Jiao Tong University</i>                                |
| 2023 年 12 月 | Weighted PROP Allocations with Subsidy<br><i>Workshop on Theoretical Computer Science, University of Macau</i>                           |
| 2023 年 12 月 | Recent Development of Weighted Fair Allocation<br><i>Invited Talk, Wuhan University</i>                                                  |
| 2023 年 8 月  | Weighted EF1 Allocations for Indivisible Chores<br><i>Young PhD Forum, IJTCS-FAW, Macao</i>                                              |
| 2023 年 7 月  | Weighted EF1 Allocations for Indivisible Chores<br><i>ACM EC, London</i>                                                                 |
| 2022 年 11 月 | Approximately EFX Allocations for Indivisible Chores<br><i>IJCAI-China, Shenzhen</i>                                                     |

## 学术服务

|            |                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------|
| 程序委员会委员    | AAMAS (2026), EC (2026), IJCAI (2026), WINE (2026), WINE (2025)                                             |
| 会议 (子) 审稿人 | SODA (2026), NeurIPS (2026), WWW (2025), ISAAC (2024), SODA (2023)                                          |
| 期刊审稿人      | Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS), SIAM Journal on Discrete Mathematics (SIDMA) |
| 组织委员会      | IJTCS-FAW (2023)                                                                                            |